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SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning
SLOT: D

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%

Duration: 90 minutes

Assessment Tool Weightage: 30%

Mark : 25

Code of Conduct:

I **Ch. Venkata Lakshmi Hymavathi (192425397)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ch. Hymavathi

Name of the student:

Ch. Venkata Lakshmi Hymavathi

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Total						

Select and review a recent research article related to the below **topic**, such as **Transfer Learning, DenseNet, PixelNet, RNN, Bidirectional RNN, Encoder–Decoder/Seq2Seq, BPTT, or LSTM**. Summarize the **problem addressed, dataset used, proposed deep learning model, methodology, performance metrics, major results, and key findings**, and critically comment on the **advantages, limitations, and possible future improvements** of the approach presented in the article.

Prepare a structured Article Review addressing the following sections:

1. Article Details

Provide the following information:

Title of the research article

Names of the authors

Journal/Conference name

Year of publication

DOI or publication link

Topic/Deep Learning technique addressed

Answer:

Title: *A Deep Bidirectional LSTM Model Enhanced by Transfer-Learning-Based Feature Extraction for Dynamic Human Activity Recognition*

Authors: Najmul Hassan, Abu Saleh Musa Miah, Jungpil Shin

Journal: Applied Sciences

Year: 2024

Volume/Issue: Volume 14, Issue 2

Article Number: 603

DOI: 10.3390/app14020603

Deep Learning Technique: MobileNetV2 + Deep Bidirectional LSTM (Deep BiLSTM) with Transfer Learning.

The article was published on **10 January 2024** and focuses on dynamic human activity recognition from video data.

2. Research Problem

What problem does the article attempt to solve?

Why is the problem important?

What are the limitations of existing approaches identified by the authors?

Answer:

Human Activity Recognition (HAR) is the process of identifying human actions from video sequences. It is important in applications such as:

- Intelligent video surveillance
- Abnormal activity detection
- Healthcare monitoring
- Video retrieval
- Security systems

Traditional approaches often have difficulty extracting both **spatial and temporal information** from videos. A single video frame cannot completely describe an activity because actions involve movement over time.

Existing methods also face problems such as:

- Large video datasets
- Similar-looking human activities

- Complex backgrounds
- Changes in illumination

3. Objectives of the Study

What are the main objectives of the research?

What specific goals does the proposed model attempt to achieve?

Answer:

The main objectives are:

1. To develop an effective deep learning model for human activity recognition.
2. To extract important spatial features from video frames using **MobileNetV2**.
3. To use **Deep BiLSTM** to learn temporal dependencies.
4. To improve recognition accuracy using transfer learning.
5. To evaluate the proposed model on multiple benchmark datasets.

4. Dataset Description

What dataset(s) are used?

What is the size of the dataset?

What type of data is used—images, text, time-series, video, etc.?

How is the dataset divided into training, validation, and testing sets?

Answer:

The researchers used three benchmark human-action datasets.

A. UCF11 / YouTube Actions Dataset

- Contains **11 activity categories**.
- Examples include:
 - Basketball
 - Biking
 - Diving
 - Golf swing
 - Horse riding
 - Soccer juggling
- Videos contain variations in camera movement, background, illumination and viewpoint.

B. JHMDB Dataset

- Contains **21 activity categories**.
- Contains **923 videos**.
- Includes activities such as catching, clapping, jumping, brushing hair, baseball swinging and shooting.

C. UCF Sports Dataset

- Contains **150 video sequences**.
- Video resolution: **720 × 480**.
- Contains sports activities such as diving, horse riding, golf swing, skateboarding and weightlifting.

The article does not clearly provide a conventional single train/validation/test percentage split in the sections available for this review, so a specific percentage should not be claimed.

5. Proposed Methodology

Explain the overall methodology adopted in the article.

Identify the deep learning architecture used.

Explain the important layers/components of the model.

Describe the data preprocessing techniques used.

Explain the training procedure and important parameters, wherever available.

Answer:

The proposed system follows these main steps:

1. Input video is collected.

2. Video is divided into individual frames.
3. **MobileNetV2** is used for feature extraction.
4. Extracted features are passed to the **Deep BiLSTM** network.
5. BiLSTM processes temporal information in both forward and backward directions.
6. The learned representation is passed to the classification layer.
7. **Softmax** is used for final activity classification.
8. The model is evaluated using benchmark datasets.

6. Deep Learning Technique Used

Explain how the selected technique is applied in the research.

How is Transfer Learning used?

How does DenseNet perform feature reuse?

How is RNN/LSTM used to process sequential data?

How does a Bidirectional RNN utilize information from both directions?

How does an Encoder–Decoder/Seq2Seq architecture process input and output sequences?

How is BPTT used for training recurrent networks?

Answer:

Deep Bidirectional LSTM

LSTM is a type of recurrent neural network designed to learn dependencies in sequential data.

An LSTM uses memory mechanisms to retain important information over time.

For video activity recognition:

Video frames → Features → Temporal sequence → LSTM → Activity

Why BiLSTM?

A normal LSTM processes information mainly in one temporal direction.

A **Bidirectional LSTM** processes the sequence in both directions:

- Forward direction
- Backward direction

Therefore, the model can use information from both past and future frames when recognizing an activity.

The proposed model uses **Deep BiLSTM** to refine MobileNetV2 features and learn temporal dependencies in video sequences.

7. Model Architecture

Draw or reproduce a simplified architecture diagram of the proposed model and explain:

Input → Preprocessing → Feature Extraction/Encoder → Hidden Layers →

Classification/Decoder → Output

Identify the role of the major components in the architecture.

Answer:

Simplified Architecture

Input Video



Video Frames



Preprocessing



MobileNetV2

(Feature Extraction)



Deep BiLSTM

(Temporal Feature Learning)



Dense Layer

↓
Softmax Classifier

↓
Predicted Activity

Role of Components

Input Video: Provides the sequence of human activities.

Preprocessing: Converts video information into suitable frame-based input.

MobileNetV2: Extracts high-level spatial features from individual video frames.

Deep BiLSTM: Learns temporal relationships between consecutive frames and processes information in both directions.

Dense Layer: Maps learned features to activity classes.

Softmax: Produces the final activity classification.

The architecture combines CNN-based spatial feature extraction with recurrent temporal modeling.

8. Performance Evaluation

Identify the evaluation metrics used in the article, such as:

Accuracy, Precision, Recall, F1-Score, Loss, Mean Squared Error (MSE)

BLEU Score, where applicable

Report the major performance results obtained by the proposed model.

Answer:

The researchers primarily evaluated the model using:

- **Accuracy**
- **Confusion Matrix**
- Accuracy/loss curves

The proposed model achieved:

Dataset	Proposed Model Accuracy
UCF11	99.20%
UCF Sports	93.30%
JHMDB	76.30%

These are the major reported results of the study.

9. Results and Discussion

What are the major findings of the research?

How does the proposed model perform compared with existing methods?

Which model/technique produced the best results?

What factors contributed to the observed performance?

Answer:

UCF11

The proposed method achieved **99.2% accuracy**.

It performed better than several compared approaches, including:

- Single-stream CNN – 93.1%
- BT-LSTM – 85.3%
- 3DCNN – 85.2%
- Deep Autoencoder + CNN – 96.2%
- ViT-ReT – 92.4%

The proposed model therefore showed strong performance on UCF11.

UCF Sports

The proposed model achieved **93.3% accuracy**.

Compared with:

- QST-CNN-LSTM – 93.2%
- GMM + KF + GRNN – 89.1%

- Two-stream LSTM – 92.2%

The proposed method provided slightly better performance.

JHMDB

The JHMDB dataset was more challenging.

The model achieved approximately **76.3–76.6% accuracy** as reported in different portions of the article. The lower performance indicates that the dataset contains more difficult activities and greater visual similarity between actions.

10. Advantages of the Proposed Model

Identify at least three advantages of the proposed approach.

Answer:

1. High Accuracy

The model achieved **99.2% accuracy on UCF11**, demonstrating strong recognition capability.

2. Spatial and Temporal Learning

MobileNetV2 extracts spatial features while Deep BiLSTM learns temporal dependencies.

3. Transfer Learning

Using a pre-trained MobileNetV2 reduces the need to learn all visual features from scratch.

4. Bidirectional Processing

BiLSTM can utilize information from both directions of a sequence.

5. Multiple Dataset Evaluation

Testing on UCF11, UCF Sports and JHMDB provides evidence of performance across different activity-recognition scenarios.

11. Limitations of the Study

Identify the major limitations mentioned by the authors and any additional limitations you observe, such as: Small dataset, High computational requirements, Training time, Overfitting, Limited generalization, Model complexity.

Answer:

The major limitations include:

1. **High computational requirements** because video-based deep learning is computationally expensive.
2. **Complex video data** can make recognition difficult.
3. **Similar activities** may cause classification errors.
4. Performance decreases on challenging datasets such as JHMDB.
5. The model uses a combination of MobileNetV2 and Deep BiLSTM, which increases architectural complexity.

12. Critical Analysis

Provide your own analysis of the research article:

Is the selected deep learning architecture appropriate for the problem?

Is the dataset adequate?

Are the evaluation metrics appropriate?

Are the experimental results convincing?

Could another Unit IV technique provide better performance?

Answer:

Is the architecture appropriate?

Yes.

The architecture is appropriate because human activities contain both spatial and temporal information. MobileNetV2 is suitable for spatial feature extraction, while BiLSTM is suitable for sequential temporal information.

Is the dataset adequate?

The use of three benchmark datasets is a strong point because it allows evaluation under different activity-recognition conditions. However, the datasets remain relatively limited compared with very large modern video datasets.

Are the evaluation metrics appropriate?

Accuracy and confusion matrices are suitable for multi-class activity recognition. However, **precision, recall and F1-score** could provide additional information, particularly for classes with imbalanced performance.

Are the results convincing?

The results are reasonably convincing because the proposed model was tested on three datasets and compared with existing approaches. The very high UCF11 accuracy demonstrates the effectiveness of the approach. However, the lower JHMDB result shows that generalization to difficult activity datasets remains a challenge.

Could another technique perform better?

A modern **Transformer-based video model** could potentially provide better long-range temporal modeling. However, it may require more computational resources and larger datasets.

13. Future Scope

Suggest possible improvements or future research directions based on the limitations identified in the article.

Answer:

Possible future improvements include:

1. Use larger and more diverse video datasets.
2. Integrate Transformer-based temporal modeling.
3. Use attention mechanisms to focus on important frames.
4. Improve recognition of similar activities.
5. Reduce computational complexity.
6. Develop lightweight versions for real-time surveillance.
7. Evaluate precision, recall and F1-score in addition to accuracy.
8. Test the model in real-world surveillance environments.
9. Investigate stronger data augmentation techniques.

14. Conclusion

Provide a brief conclusion summarizing:

The research problem

Proposed model

Major results

Your overall assessment of the article

Answer:

The reviewed article presents a **Deep Bidirectional LSTM model enhanced with transfer-learning-based MobileNetV2 feature extraction** for dynamic human activity recognition. The model combines the strengths of CNN and BiLSTM. MobileNetV2 extracts spatial features from video frames, while Deep BiLSTM learns temporal relationships in the sequence.

The proposed approach achieved **99.20% accuracy on UCF11, 93.30% on UCF Sports and approximately 76.3% on JHMDB**.

Overall, the architecture is well suited to video-based human activity recognition because it considers both spatial and temporal information. Its major strengths are high accuracy, transfer learning and bidirectional temporal processing. **Overall assessment:** The article provides a strong and practical application of **LSTM, BiLSTM and Transfer Learning** for human activity recognition and is suitable for demonstrating modern deep learning techniques.

15. References

Provide the complete citation of the selected research article and any additional references used in the review.

Answer:

Hassan, N., Miah, A. S. M., & Shin, J. (2024). *A Deep Bidirectional LSTM Model Enhanced by Transfer-Learning-Based Feature Extraction for Dynamic Human Activity Recognition*. *Applied Sciences*, 14(2), 603. <https://doi.org/10.3390/app14020603>.

Additional references mentioned in the article:

1. UCF11 / YouTube Actions Dataset.
2. UCF Sports Dataset.
3. JHMDB Dataset.
4. MobileNetV2 architecture.
5. Long Short-Term Memory (LSTM) networks.
6. Bidirectional LSTM networks.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis